

(bars: lengths by R , gain by a_{nom} ; hats on estimates; $e_\theta = \theta - \hat{\theta}$; posterior quantities unless marked $-$; $\bar{w}_d[k]$ commanded height, $\Delta \bar{w}_d[k] = \bar{w}_d[k+1] - \bar{w}_d[k]$; $\bar{a}' := \bar{a}'_{\text{true}}(\bar{w}_d[k] - \delta \hat{w}_3[k])$, $\bar{a}'' := \bar{a}''_{\text{true}}(\bar{w}_d[k] - \delta \hat{w}_3[k])$, both exogenous; $\Delta \bar{w}[k] := \bar{w}[k+1] - \bar{w}[k]$; $\Delta \hat{w}[k] := \Delta \bar{w}_d[k] + (1 - \lambda_c)[\delta \hat{w}_3[k] + y_1[k] - \delta \hat{w}_1[k]]$; $e_{\Delta \bar{w}} := \Delta \bar{w} - \Delta \hat{w}$; $\mathbb{E}[\cdot]$ mean over noise realizations conditional on the data up to k ; $P_{ij} = \mathbb{E}[e_i e_j]$; $d = 2$)

1 True dynamics

$$\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k] \\
\delta \bar{w}_3[k+1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta \bar{w}[k] &= \Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\bar{a}_w[k+1] - \bar{a}_w[k] &= \bar{a}'_{\text{true}}(\bar{w}[k]) \Delta \bar{w}[k] + \frac{1}{2} \bar{a}''_{\text{true}}(\bar{w}[k]) \Delta \bar{w}[k]^2 + \mathcal{O}(\Delta \bar{w}^3) \\
\bar{a}'_{\text{true}}(\bar{w}[k]) &= \bar{a}'_{\text{true}}(\bar{w}_d[k] - \delta \hat{w}_3[k] - e_{\delta \bar{w}_3}[k]) = \bar{a}' - \bar{a}'' e_{\delta \bar{w}_3}[k] + \mathcal{O}(e^2) \\
\bar{a}''_{\text{true}}(\bar{w}[k]) &= \bar{a}'' + \mathcal{O}(e)
\end{aligned}$$

$$\bar{a}_w[k+1] - \bar{a}_w[k] = (\bar{a}' - \bar{a}'' e_{\delta \bar{w}_3}[k]) \Delta \bar{w}[k] + \frac{1}{2} \bar{a}'' \Delta \bar{w}[k]^2$$

$$\begin{aligned}
F_{dw}[k] &= \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k-i] \\
\varepsilon_w[k] &= \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k-i] \bar{f}_T[k-i] + (1 - \lambda_c) n_w[k] \\
y_1[k] &= \delta \bar{w}_1[k] + n_w[k]
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}[\varepsilon_w[k] n_w[k]] &= (1 - \lambda_c) R_1 \\
\tilde{\varepsilon}_w[k] &:= \varepsilon_w[k] - (1 - \lambda_c) n_w[k], \quad \mathbb{E}[\tilde{\varepsilon}_w n_w] = 0, \quad \mathbb{E}[\tilde{\varepsilon}_w^2] = Q_{33} - (1 - \lambda_c)^2 R_1 =: \tilde{Q}_{33} \\
\Delta \bar{w}[k] &= \Delta \bar{w}_d[k] + (1 - \lambda_c) [\delta \bar{w}_3[k] + n_w[k]] + F_{dw}[k] e_{\bar{a}_w}[k] + \tilde{\varepsilon}_w[k]
\end{aligned}$$

2 Estimator

$$\begin{aligned}
\delta \hat{w}_3[k+1] &= \lambda_c \delta \hat{w}_3[k] - (1 - \lambda_c) [y_1[k] - \delta \hat{w}_1[k]] \\
\Delta \hat{w}[k] &= \Delta \bar{w}_d[k] + (1 - \lambda_c) [\delta \hat{w}_3[k] + y_1[k] - \delta \hat{w}_1[k]] \\
\hat{a}_w[k+1] - \hat{a}_w[k] &= \bar{a}' \Delta \hat{w}[k] + \frac{1}{2} \bar{a}'' \Delta \hat{w}[k]^2
\end{aligned}$$

3 Error increment

$$y_1[k] - \delta \hat{w}_1[k] = e_{\delta \bar{w}_1}[k] + n_w[k]$$

$$e_{\Delta \bar{w}}[k] = \Delta \bar{w}[k] - \Delta \hat{w}[k] = (1 - \lambda_c)[e_{\delta \bar{w}_3}[k] - e_{\delta \bar{w}_1}[k]] + F_{dw}[k] e_{\bar{a}_w}[k] + \tilde{\varepsilon}_w[k]$$

$$e_{\bar{a}_w}[k+1] - e_{\bar{a}_w}[k] = (\bar{a}_w[k+1] - \bar{a}_w[k]) - (\hat{a}_w[k+1] - \hat{a}_w[k])$$

$$= (\bar{a}' - \bar{a}'' e_{\delta \bar{w}_3})(\Delta \hat{w} + e_{\Delta \bar{w}}) + \frac{1}{2} \bar{a}'' (\Delta \hat{w} + e_{\Delta \bar{w}})^2 - \bar{a}' \Delta \hat{w} - \frac{1}{2} \bar{a}'' \Delta \hat{w}^2$$

$$= \bar{a}' e_{\Delta \bar{w}} - \bar{a}'' e_{\delta \bar{w}_3} \Delta \hat{w} - \bar{a}'' e_{\delta \bar{w}_3} e_{\Delta \bar{w}} + \bar{a}'' \Delta \hat{w} e_{\Delta \bar{w}} + \frac{1}{2} \bar{a}'' e_{\Delta \bar{w}}^2 + \mathcal{O}(3)$$

4 Mean

$$\mathbb{E}[e_{\delta \bar{w}_1}] = \mathbb{E}[e_{\delta \bar{w}_3}] = \mathbb{E}[e_{\bar{a}_w}] = \mathbb{E}[\tilde{\varepsilon}_w] = 0$$

$$\mathbb{E}[e_{\delta \bar{w}_3}^2] = P_{33}, \quad \mathbb{E}[e_{\delta \bar{w}_1}^2] = P_{11}, \quad \mathbb{E}[e_{\delta \bar{w}_1} e_{\delta \bar{w}_3}] = P_{13}$$

$$\mathbb{E}[e_{\delta \bar{w}_3} e_{\bar{a}_w}] = P_{34}, \quad \mathbb{E}[e_{\delta \bar{w}_1} e_{\bar{a}_w}] = P_{14}, \quad \mathbb{E}[e_{\bar{a}_w}^2] = P_{44}$$

$$\mathbb{E}[\tilde{\varepsilon}_w e_{\delta \bar{w}_1}] = \mathbb{E}[\tilde{\varepsilon}_w e_{\delta \bar{w}_3}] = \mathbb{E}[\tilde{\varepsilon}_w e_{\bar{a}_w}] = \mathbb{E}[\tilde{\varepsilon}_w n_w] = 0$$

$$\mathbb{E}[e_j \delta \hat{w}_i] = \mathbb{E}[e_j y_1] = 0 \quad (\text{posterior error} \perp \text{data})$$

$$\mathbb{E}[\bar{a}' e_{\Delta \bar{w}}] = 0$$

$$\mathbb{E}[-\bar{a}'' e_{\delta \bar{w}_3} \Delta \hat{w}] = 0$$

$$\mathbb{E}[-\bar{a}'' e_{\delta \bar{w}_3} e_{\Delta \bar{w}}] = -\bar{a}'' [(1 - \lambda_c)(P_{33} - P_{13}) + F_{dw} P_{34}]$$

$$\mathbb{E}[\bar{a}'' \Delta \hat{w} e_{\Delta \bar{w}}] = 0$$

$$\mathbb{E}[\frac{1}{2} \bar{a}'' e_{\Delta \bar{w}}^2] = \frac{1}{2} \bar{a}'' [(1 - \lambda_c)^2 (P_{33} + P_{11} - 2P_{13}) + 2(1 - \lambda_c) F_{dw} (P_{34} - P_{14})$$

$$+ F_{dw}^2 P_{44} + \tilde{Q}_{33}]$$

$$\mathbb{E}[e_{\bar{a}_w}[k+1]] - \mathbb{E}[e_{\bar{a}_w}[k]] = \bar{a}'' \left\{ -(1 - \lambda_c)(P_{33} - P_{13}) - F_{dw} P_{34} \right.$$

$$+ \frac{1}{2} (1 - \lambda_c)^2 (P_{33} + P_{11} - 2P_{13}) + (1 - \lambda_c) F_{dw} (P_{34} - P_{14})$$

$$\left. + \frac{1}{2} F_{dw}^2 P_{44} + \frac{1}{2} \tilde{Q}_{33} \right\}$$

(first, second and fourth lines: $\bar{a}'$ and $\Delta \hat{w}$ are functions of the data, $e_{\Delta \bar{w}}$ and $e_{\delta \bar{w}_3}$ are posterior errors, $\mathbb{E}[e | \text{data}] = 0$; without the $y_1 - \delta \hat{w}_1$ input $e_{\Delta \bar{w}}$ contains $(1 - \lambda_c)n_w$, which is correlated with the data, and these means become the n_w feedthrough shares of 0902_formC_aptrue_4state.tex, Error dynamics, each $\propto \bar{a}''(1 - \lambda_c)\ell_{31}R_1$, whose sum was measured to vanish)

5 Estimator with the mean term

$$\begin{aligned}\hat{a}_w[k+1] &= \hat{a}_w[k] + \bar{a}' \Delta \hat{w}[k] + \frac{1}{2} \bar{a}'' \Delta \hat{w}[k]^2 \\ &\quad + \bar{a}'' \left\{ -(1 - \lambda_c)(P_{33}[k] - P_{13}[k]) - F_{dw}[k] P_{34}[k] \right. \\ &\quad \left. + \frac{1}{2}(1 - \lambda_c)^2(P_{33}[k] + P_{11}[k] - 2P_{13}[k]) \right. \\ &\quad \left. + (1 - \lambda_c)F_{dw}[k](P_{34}[k] - P_{14}[k]) + \frac{1}{2}F_{dw}[k]^2 P_{44}[k] + \frac{1}{2}\tilde{Q}_{33}[k] \right\}\end{aligned}$$

$$\mathbb{E}[e_{\bar{a}_w}[k+1]] - \mathbb{E}[e_{\bar{a}_w}[k]] = 0$$

(the braced term is a deterministic input to the predict; it is not differentiated and does not enter F_e , H , Q)

6 Jacobians

$$\begin{aligned}\frac{\partial \delta \hat{w}_3[k+1]}{\partial \delta \hat{w}_3[k]} &= \lambda_c, & \frac{\partial \delta \hat{w}_3[k+1]}{\partial \delta \hat{w}_1[k]} &= (1 - \lambda_c) \\ \frac{\partial \hat{a}_w[k+1]}{\partial \delta \hat{w}_3[k]} &= (1 - \lambda_c)(\bar{a}' + \bar{a}'' \Delta \hat{w}[k]) - \bar{a}'' \Delta \hat{w}[k] \rightarrow (1 - \lambda_c) \bar{a}' \\ \frac{\partial \hat{a}_w[k+1]}{\partial \delta \hat{w}_1[k]} &= -(1 - \lambda_c)(\bar{a}' + \bar{a}'' \Delta \hat{w}[k]) \rightarrow -(1 - \lambda_c) \bar{a}' \\ \frac{\partial \bar{a}_w[k+1]}{\partial \bar{a}_w[k]} &= 1 + F_{dw}[k](\bar{a}' + \bar{a}'' \Delta \hat{w}[k]) \rightarrow 1 + F_{dw}[k] \bar{a}'\end{aligned}$$

(limits at $\Delta \hat{w} \rightarrow 0$, the tangent the code keeps; the $-\bar{a}'' \Delta \hat{w}$ in the first gain-row entry is $\partial \bar{a}' / \partial \delta \hat{w}_3 = -\bar{a}''$ of the 'est' reading; the last entry and $\partial \delta \bar{w}_3[k+1] / \partial \bar{a}_w[k] = -F_{dw}$ are entries of the TRUE transition, not of the estimator's lines (which do not contain $\hat{a}_w$): the truth's $F_{dw} e_{\bar{a}_w}$ inside $\Delta \bar{w}$, written as $F_{dw}(\bar{a}_w - \hat{a}_w)$)

7 Error dynamics

$$\begin{aligned}e_{\delta \bar{w}_1}[k+1] &= e_{\delta \bar{w}_2}[k] \\ e_{\delta \bar{w}_2}[k+1] &= e_{\delta \bar{w}_3}[k] \\ e_{\delta \bar{w}_3}[k+1] &= \lambda_c e_{\delta \bar{w}_3}[k] + (1 - \lambda_c) e_{\delta \bar{w}_1}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \tilde{\varepsilon}_w[k] \\ e_{\bar{a}_w}[k+1] &= -(1 - \lambda_c) \bar{a}' e_{\delta \bar{w}_1}[k] + (1 - \lambda_c) \bar{a}' e_{\delta \bar{w}_3}[k] + (1 + F_{dw} \bar{a}') e_{\bar{a}_w}[k] + \bar{a}' \tilde{\varepsilon}_w[k] \\ &\quad - \bar{a}'' e_{\delta \bar{w}_3} \Delta \hat{w} - \bar{a}'' e_{\delta \bar{w}_3} e_{\Delta \bar{w}} + \bar{a}'' \Delta \hat{w} e_{\Delta \bar{w}} + \frac{1}{2} \bar{a}'' e_{\Delta \bar{w}}^2 - \mathbb{E}[\cdot]\end{aligned}$$

(third line: $\delta \bar{w}_3[k+1] - \delta \hat{w}_3[k+1]$ with $y_1 - \delta \hat{w}_1 = e_{\delta \bar{w}_1} + n_w$, the n_w of ε_w cancels; fourth line: the second-order group minus its mean of the previous section)

$$\mathbf{e}[k+1] = \tilde{F}_e[k] \mathbf{e}[k] + \tilde{\mathbf{q}}[k], \quad \tilde{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ (1-\lambda_c) & 0 & \lambda_c & -F_{dw} \\ -(1-\lambda_c)\bar{a}' & 0 & (1-\lambda_c)\bar{a}' & 1+F_{dw}\bar{a}' \end{bmatrix}$$

$$\tilde{F}_e[k] = F_e[k] - g_n[k] \mathbf{e}_1^\top, \quad F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} \\ 0 & 0 & (1-\lambda_c)\bar{a}' & 1+F_{dw}\bar{a}' \end{bmatrix}$$

$$g_n[k] = (1-\lambda_c) \begin{bmatrix} 0 \\ 0 \\ -1 \\ \bar{a}' \end{bmatrix}, \quad \tilde{\mathbf{q}}[k] = \tilde{\varepsilon}_w[k] \begin{bmatrix} 0 \\ 0 \\ -1 \\ \bar{a}' \end{bmatrix}, \quad \mathbb{E}[\tilde{\mathbf{q}}[k] n_w[k]] = 0$$

$$P^f[k+1] = \tilde{F}_e[k] P[k] \tilde{F}_e[k]^\top + \tilde{Q}[k]$$

8 Measurement and update

$$\nabla_d \bar{w}_d[k] = \bar{w}_d[k] - \bar{w}_d[k-d]$$

$$y_1[k] = \delta \bar{w}_1[k] + n_w[k]$$

$$y_2[k] = \bar{a}_w[k-d] + n_a[k], \quad \bar{a}_w[k-d] = \bar{a}_w[k] - \bar{a}' \nabla_d \bar{w}_d[k]$$

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$e_{y_1}[k] = e_{\delta \bar{w}_1}^-[k] + n_w[k], \quad e_{y_2}[k] = e_{\bar{a}_w}^-[k] + n_a[k]$$

$$L[k] = P^f H^\top (H P^f H^\top + R)^{-1}$$

$$\hat{\mathbf{x}}[k] = \hat{\mathbf{x}}^-[k] + L[k] \mathbf{e}_y[k]$$

$$P[k] = (I - L[k] H) P^f[k]$$

$$y_1[k] - \delta \hat{w}_1[k] = \frac{R_1}{P_{11}^f[k] + R_1} e_{y_1}[k] - \ell_{12}[k] (e_{y_2}[k] - \ell_{41}[k] e_{y_1}[k]) = \mathbb{E}[n_w[k] | y[1..k]]$$

(last line: the posterior residual the Estimator section feeds into the next step, written for the sequential update y_1 then y_2 as in the code; $e_{y_2} - \ell_{41} e_{y_1}$ is the y_2 innovation after the y_1 update, $\ell_{12} = H_{24} P_{14}^{(1)} / S_2$ equals the joint L_{12} ; exact in the linear-Gaussian model, approximate for the EKF)

9 Noise

$$\begin{aligned}
\kappa_T &= \frac{4k_B T a_o}{R}, & \sigma_{n_w}^2 &= \frac{\sigma_n^2}{R^2} \\
\tilde{Q}_{33} &= \kappa_T \left(\hat{a}_w[k] + (1 - \lambda_c)^2 (\hat{a}_w[k-1] + \hat{a}_w[k-2]) \right) & & (= Q_{33} - (1 - \lambda_c)^2 \sigma_{n_w}^2) \\
\tilde{Q}_{34} &= \tilde{Q}_{43} = -\bar{a}' \tilde{Q}_{33} \\
\tilde{Q}_{44} &= \bar{a}'^2 \tilde{Q}_{33} \\
R &= \text{diag}(R_1, R_2), & R_1 &= \sigma_{n_w}^2, & R_2 &= 2a_{\text{cov}}^2 IF_{\text{var}}(\hat{a}_w) (\hat{a}_w + \xi)^2
\end{aligned}$$

$(IF_{\text{var}}, \xi$ as in 0902_formC_aptrue_4state.tex; the n_w share $(1 - \lambda_c)^2 \sigma_{n_w}^2$ has left Q and entered the Estimator section as the input $g_n(y_1 - \delta \hat{w}_1)$)

10 b_{true} arm: the law read at the estimated gain

(this arm: $b := b_{\text{true}}(\bar{w}[k-1])$ read at the previous step's TRUE height (code, `b_true_at = 'true'`), treated as exogenous; slot 5 locked; $\bar{a}' := b(1 - \hat{a}_w[k])^2$ and $\bar{a}'' := -2b(1 - \hat{a}_w[k])\bar{a}'$ read at the ESTIMATED GAIN, not at a height; $\frac{\partial \bar{a}'}{\partial \hat{a}_w} = -2b(1 - \hat{a}_w) = \frac{\bar{a}''}{\bar{a}'}$; everything not listed is as in Sections 1–9)

True dynamics, the slope line

$$\bar{a}'_{\text{true}}(\bar{w}[k]) = b(1 - \bar{a}_w[k])^2 = b(1 - \hat{a}_w[k] - e_{\bar{a}_w}[k])^2 = \bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w}[k] + \mathcal{O}(e^2)$$

$$\bar{a}_w[k+1] - \bar{a}_w[k] = \left(\bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w}[k]\right) \Delta \bar{w}[k] + \frac{1}{2} \bar{a}'' \Delta \bar{w}[k]^2$$

Error increment (the $-\bar{a}'' e_{\delta \bar{w}_3}$ lines of Section 3 are replaced)

$$e_{\bar{a}_w}[k+1] - e_{\bar{a}_w}[k] = \bar{a}' e_{\Delta \bar{w}} + \frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w} \Delta \hat{w} + \frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w} e_{\Delta \bar{w}} + \bar{a}'' \Delta \hat{w} e_{\Delta \bar{w}} + \frac{1}{2} \bar{a}'' e_{\Delta \bar{w}}^2 + \mathcal{O}(3)$$

Mean

$$\begin{aligned} \mathbb{E}\left[\frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w} \Delta \hat{w}\right] &= 0 \\ \mathbb{E}\left[\frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w} e_{\Delta \bar{w}}\right] &= \frac{\partial \bar{a}'}{\partial \hat{a}_w} [(1 - \lambda_c)(P_{34} - P_{14}) + F_{dw}P_{44}] \\ \mathbb{E}[e_{\bar{a}_w}[k+1]] - \mathbb{E}[e_{\bar{a}_w}[k]] &= \frac{\partial \bar{a}'}{\partial \hat{a}_w} [(1 - \lambda_c)(P_{34} - P_{14}) + F_{dw}P_{44}] \\ &\quad + \frac{1}{2} \bar{a}'' [(1 - \lambda_c)^2(P_{33} + P_{11} - 2P_{13}) + 2(1 - \lambda_c)F_{dw}(P_{34} - P_{14}) \\ &\quad + F_{dw}^2P_{44} + \tilde{Q}_{33}] \end{aligned}$$

Estimator with the mean term

$$\begin{aligned} \hat{a}_w[k+1] &= \hat{a}_w[k] + \bar{a}' \Delta \hat{w}[k] + \frac{1}{2} \bar{a}'' \Delta \hat{w}[k]^2 \\ &\quad + \frac{\partial \bar{a}'}{\partial \hat{a}_w} [(1 - \lambda_c)(P_{34}[k] - P_{14}[k]) + F_{dw}[k]P_{44}[k]] \\ &\quad + \frac{1}{2} \bar{a}'' [(1 - \lambda_c)^2(P_{33}[k] + P_{11}[k] - 2P_{13}[k]) + 2(1 - \lambda_c)F_{dw}[k](P_{34}[k] - P_{14}[k]) \\ &\quad + F_{dw}[k]^2P_{44}[k] + \tilde{Q}_{33}[k]] \end{aligned}$$

$$\mathbb{E}[e_{\bar{a}_w}[k+1]] - \mathbb{E}[e_{\bar{a}_w}[k]] = 0$$

Difference to the Section 5 term (gain-read minus height-read start point)

$$\begin{aligned} &\frac{\partial \bar{a}'}{\partial \hat{a}_w} [(1 - \lambda_c)(P_{34} - P_{14}) + F_{dw}P_{44}] + \bar{a}'' [(1 - \lambda_c)(P_{33} - P_{13}) + F_{dw}P_{34}] \\ &= \frac{\partial \bar{a}'}{\partial \hat{a}_w} [(1 - \lambda_c)(P_{34} + \bar{a}'P_{33} - P_{14} - \bar{a}'P_{13}) + F_{dw}(P_{44} + \bar{a}'P_{34})] \\ &= \frac{\partial \bar{a}'}{\partial \hat{a}_w} \mathbb{E}\left[(e_{\bar{a}_w} + \bar{a}' e_{\delta \bar{w}_3}) e_{\Delta \bar{w}}\right] \end{aligned}$$

(= 0 when the gain error is slaved to the position error through the law, $e_{\bar{a}_w} = -\bar{a}' e_{\delta \bar{w}_3}$, i.e. $P_{34} = -\bar{a}'P_{33}$, $P_{14} = -\bar{a}'P_{13}$ and $P_{44} = \bar{a}'^2P_{33}$ (the first two alone leave $F_{dw}(P_{44} - \bar{a}'^2P_{33})$); hold, both

trajectories: $P_{34} + \bar{a}' P_{33} \approx 3 \times 10^{-9}$, the difference is $< 0.005 \times 10^{-6}$ per step; near-wall half of the descent: -3.5×10^{-6} (canon) and -0.5×10^{-6} (Meng) per step in $\mathbb{E}[\hat{a}_w - \bar{a}_w]$, running sums -0.0026 and -0.0040 at the start of the hold (code: ctrl_const.pred_mean2_e4, the difference of the two pred_mean2 logs, 10 seeds, run_btrue_e4.m); the closed-loop response to this input is smaller than its running sum, $+0.0010$ (canon) and $+0.0004$ (Meng) at the end of the descent, relaxing to < 0.0003 within the following second)

(treating b as exogenous drops the mean $-(1 - \hat{a}_w)^2 b'_{\text{true}} \mathbb{E}[e_{\delta\bar{w}_3}[k-1] e_{\Delta\bar{w}}]$ of $\bar{a}' e_{\Delta\bar{w}}$, b being read at the true height $\bar{w}_d[k-1] - \delta\bar{w}_3[k-1]$; relative to the kept $\bar{a}''$ term it is $\sim b'_{\text{true}}/(2b^2(1 - \hat{a}_w))$, a few percent, b_{true} spanning 8/9 to 1 over the band)

Jacobians and error dynamics (the entries that change)

$$\begin{aligned} \frac{\partial \hat{a}_w[k+1]}{\partial \delta\hat{w}_3[k]} &= (1 - \lambda_c)(\bar{a}' + \bar{a}'' \Delta\hat{w}[k]) \rightarrow (1 - \lambda_c) \bar{a}' \\ \frac{\partial \hat{a}_w[k+1]}{\partial \hat{a}_w[k]} &= 1 + \frac{\partial \bar{a}'}{\partial \hat{a}_w} \Delta\hat{w}[k] + F_{dw}[k](\bar{a}' + \bar{a}'' \Delta\hat{w}[k]) \rightarrow 1 + \frac{\partial \bar{a}'}{\partial \hat{a}_w} \Delta\hat{w}[k] + F_{dw}[k] \bar{a}' \\ \tilde{F}_e(4, 4) &= 1 + F_{dw} \bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{a}_w} \Delta\hat{w} \\ y_2[k] &= \bar{a}_w[k] - \bar{a}' \nabla_d \bar{w}_d[k] + n_a[k], \quad H_{24} = 1 - \frac{\partial \bar{a}'}{\partial \hat{a}_w} \nabla_d \bar{w}_d[k] \end{aligned}$$

(no $-\bar{a}'' \Delta\hat{w}$ in the $\delta\hat{w}_3$ entry: the slope does not read a height; $H_{24} \neq 1$ whenever $\bar{a}'$ reads $\hat{a}_w$ (Sections 10, 11; code line 1 - Grad_wbar_d*A_a_i), $= 1$ only for the exogenous slope of Sections 1–9; the $\partial \bar{a}' / \partial \hat{a}_w \Delta\hat{w}$ entry is the law's own amplification of a gain error along the step, > 1 when moving toward the wall; $\tilde{Q}_{34} = -\bar{a}' \tilde{Q}_{33}$, $\tilde{Q}_{44} = \bar{a}'^2 \tilde{Q}_{33}$ unchanged)

11 Production: b a constant state

(this arm: $\hat{b} := \hat{x}_5$, constant model, $\tilde{Q}_{55} = 0$, seed $8/9$, $\sqrt{P_{55}[0]} = \sup_{\text{env}} |b_{\text{true}} - 8/9|$; $\bar{a}' := \hat{b}(1 - \hat{a}_w)^2$, $\frac{\partial \bar{a}'}{\partial \hat{b}} = (1 - \hat{a}_w)^2 = \frac{\bar{a}'}{\hat{b}}$, $\bar{a}'' := -2\hat{b}(1 - \hat{a}_w)\bar{a}'$; $e_b[k] := b_{\text{true}}(\bar{w}[k]) - \hat{b}[k]$, NOT zero-mean: b_{true} is a curve, $\hat{b}$ a constant; $f(\bar{w}) := (1 - \bar{a}_{\text{true}}(\bar{w}))^2(b_{\text{true}}(\bar{w}) - \hat{b})$, $F' = f$; everything not listed is as in Sections 1–10)

True dynamics, slope and curvature

$$\begin{aligned}\bar{a}'_{\text{true}}(\bar{w}[k]) &= b_{\text{true}}(\bar{w}[k])(1 - \bar{a}_w[k])^2 = \bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w}[k] + \frac{\partial \bar{a}'}{\partial \hat{b}} e_b[k] + \mathcal{O}(e^2) \\ \bar{a}''_{\text{true}}(\bar{w}[k]) &= \frac{d}{d\bar{w}} [b_{\text{true}}(\bar{w})(1 - \bar{a}_{\text{true}}(\bar{w}))^2] = \bar{a}'' + f'(\bar{w}[k]) + \frac{\bar{a}''}{\bar{a}'} f(\bar{w}[k]) + \mathcal{O}(e_{\bar{a}_w})\end{aligned}$$

($\bar{a}''/\bar{a}' = -2\hat{b}(1 - \hat{a}_w)$; the $(\bar{a}''/\bar{a}')f$ piece is first order in e_b , the same order as the model-error line, and is kept)

$\hat{b}$ and e_b

$$\begin{aligned}\hat{b}^-[k+1] &= \hat{b}[k] \\ \hat{b}[k+1] &= \hat{b}^-[k+1] + \ell_{51}[k+1] e_{y_1}[k+1] + \ell_{52}[k+1] e_{y_2}[k+1] \\ e_b[k+1] &= e_b[k] + b'_{\text{true}}(\bar{w}[k]) \Delta \bar{w}[k] - \ell_{51}[k+1] e_{y_1}[k+1] - \ell_{52}[k+1] e_{y_2}[k+1]\end{aligned}$$

Error increment (Section 10 plus the e_b and f' lines)

$$\begin{aligned}e_{\bar{a}_w}[k+1] - e_{\bar{a}_w}[k] &= \bar{a}' e_{\Delta \bar{w}} + \frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w}(\Delta \hat{w} + e_{\Delta \bar{w}}) + \frac{\partial \bar{a}'}{\partial \hat{b}} e_b(\Delta \hat{w} + e_{\Delta \bar{w}}) + \bar{a}'' \Delta \hat{w} e_{\Delta \bar{w}} + \frac{1}{2} \bar{a}'' e_{\Delta \bar{w}}^2 \\ &\quad + \frac{1}{2} \left(f' + \frac{\bar{a}''}{\bar{a}'} f \right) (\Delta \hat{w} + e_{\Delta \bar{w}})^2 + \mathcal{O}(3)\end{aligned}$$

Mean, the e_b and curvature lines ($P_{i5} :=$ the filter's own posterior cross-covariance under its constant- b model; the part of e_b that fluctuates given the data is $-b'_{\text{true}} e_{\delta \bar{w}_3}$, which is NOT P_{i5})

$$\begin{aligned}\mathbb{E} \left[\frac{\partial \bar{a}'}{\partial \hat{b}} e_b \Delta \hat{w} \right] &= \frac{\partial \bar{a}'}{\partial \hat{b}} \mathbb{E}[e_b] \Delta \hat{w} \quad (\text{first order; } \mathbb{E}[e_b] = b_{\text{true}}(\bar{w}) - \mathbb{E}[\hat{b}] \neq 0) \\ \mathbb{E} \left[\frac{\partial \bar{a}'}{\partial \hat{b}} e_b e_{\Delta \bar{w}} \right] &= \frac{\partial \bar{a}'}{\partial \hat{b}} [(1 - \lambda_c)(P_{35} - P_{15}) + F_{dw} P_{45}] \\ &\quad - \frac{\partial \bar{a}'}{\partial \hat{b}} b'_{\text{true}} [(1 - \lambda_c)(P_{33} - P_{13}) + F_{dw} P_{34}] \\ \mathbb{E} \left[\frac{1}{2} \left(f' + \frac{\bar{a}''}{\bar{a}'} f \right) (\Delta \hat{w} + e_{\Delta \bar{w}})^2 \right] &= \frac{1}{2} \left(f' + \frac{\bar{a}''}{\bar{a}'} f \right) (\Delta \hat{w}^2 + \mathbb{E}[e_{\Delta \bar{w}}^2])\end{aligned}$$

(second line: the first bracket is the filter's cross-covariance with its constant unknown b , the second is the height dependence of b_{true} through $e_{\delta \bar{w}_3}$; b'_{true} is not c -free)

$$\begin{aligned}\mathbb{E}[e_{\bar{a}_w}[k+1]] - \mathbb{E}[e_{\bar{a}_w}[k]] &= \frac{\partial \bar{a}'}{\partial \hat{b}} \mathbb{E}[e_b] \Delta \hat{w} + \frac{1}{2} \left(f' + \frac{\bar{a}''}{\bar{a}'} f \right) (\Delta \hat{w}^2 + \mathbb{E}[e_{\Delta \bar{w}}^2]) \\ &\quad - \frac{\partial \bar{a}'}{\partial \hat{b}} b'_{\text{true}} [(1 - \lambda_c)(P_{33} - P_{13}) + F_{dw} P_{34}] \\ &\quad + \frac{\partial \bar{a}'}{\partial \hat{b}} [(1 - \lambda_c)(P_{35} - P_{15}) + F_{dw} P_{45}] + (\text{Section 10 lines})\end{aligned}$$

(first two lines: model error, needs b_{true} and b'_{true} , not c -free, not added; third line: added, the Section 10 term with e_b)

Estimator with the mean term (the line added to Section 10's)

$$\hat{a}_w[k+1] = (\text{Section 10}) + \frac{\partial \bar{a}'}{\partial \hat{b}} [(1 - \lambda_c)(P_{35}[k] - P_{15}[k]) + F_{dw}[k] P_{45}[k]]$$

Model error along a path (first order in e_b ; open loop, $\mu := \mathbb{E}[e_{\bar{a}_w}]_{\text{model}}$)

$$\begin{aligned} \mu[k+1] &= \left(1 + \frac{\partial \bar{a}'}{\partial \hat{a}_w} \Delta \bar{w}[k]\right) \mu[k] + f(\bar{w}[k]) \Delta \bar{w}[k] + \frac{1}{2} \left(f' + \frac{\bar{a}''}{\bar{a}'} f\right) \Delta \bar{w}[k]^2 + \mathcal{O}(3) \\ \frac{d\mu}{d\bar{w}} &= \frac{\partial \bar{a}'}{\partial \hat{a}_w} \mu + f \quad \implies \quad \mu(\bar{w}) = (1 - \bar{a}_{\text{true}}(\bar{w}))^2 \int_{\bar{w}_{\text{start}}}^{\bar{w}} (b_{\text{true}} - \hat{b}) d\bar{w}' \\ \frac{1}{1 - \bar{a}_{\text{true}}(\bar{w})} - \frac{1}{1 - \hat{a}_w(\bar{w})} &= \int_{\bar{w}_{\text{start}}}^{\bar{w}} (b_{\text{true}} - \hat{b}) d\bar{w}' \quad (\text{exact, noise-free: } \frac{d}{d\bar{w}} \frac{1}{1 - \bar{a}} = b) \\ \text{stationary hold: } \mathbb{E}[\Delta \mu] &\rightarrow 0 \\ \mathbb{E}[f \Delta \bar{w}] &= -(1 - \lambda_c) f' \text{Var}(\bar{w}), \quad \frac{1}{2} f' \mathbb{E}[\Delta \bar{w}^2] = +(1 - \lambda_c) f' \text{Var}(\bar{w}) \\ \frac{1}{2} \frac{\bar{a}''}{\bar{a}'} f \mathbb{E}[\Delta \bar{w}^2] + \frac{\partial \bar{a}'}{\partial \hat{a}_w} \mathbb{E}[\Delta \bar{w} e_{\bar{a}_w}] &= 0 \quad (\text{likewise}) \end{aligned}$$

(the forcing f alone integrates to $F(\bar{w}_{\text{end}}) - F(\bar{w}_{\text{start}})$, $F' = f$; the law then AMPLIFIES what was injected far away by $[(1 - \bar{a}_{\text{wall}})/(1 - \bar{a}_{\text{far}})]^2 = 29.8$ from $\bar{w} = 6.67$ to 1.10 – the open loop of the measured $\times 22$)

($\hat{b} = 8/9$, $c_{\perp}$ of the plant: $b_{\text{true}}(6.67) = 0.883$, minimum 0.867 at $\bar{w} = 2.19$, crosses $8/9$ at $\bar{w} = 1.325$, $b_{\text{true}}(1.10) = 0.928$; $f(1.10) = +0.032$, $f(1.8) = -0.0066$, $f(6.67) = -0.0002$. Open loop, $\bar{a}_w - \hat{a}_w$ at the wall: Meng starts at $\bar{w} = 6.67$ ($15 \mu\text{m}$): $\int_{6.67}^{1.10} (b_{\text{true}} - 8/9) d\bar{w} = +0.068$, $\mu(1.10) = (1 - 0.0873)^2 \times 0.068 = +0.057$ (exact law integration $+0.060$, 69% of $\bar{a}_w(1.10)$); canon starts at $\bar{w} = 22.2$ ($50 \mu\text{m}$): $+0.084$ ($+0.093$ exact). The estimate arrives BELOW the truth, the long stretch with $b_{\text{true}} < 8/9$ outweighing the short one above it; the 09-04 first writing integrated f alone and gave 0.008 – FAIL caught by the independent verification. MEASURED (run_prod_ladder.m, four blocks, 10 seeds, paired to the b_{true} -curve arm): b locked at $8/9$: end of descent -0.0030 ± 0.0002 (canon), -0.0005 ± 0.0002 (Meng); hold start -0.0005 , -0.0009 ; the y_2 leg in motion removes $\geq 95\%$ of the open-loop error, sign as derived. b locked at $b_{\text{true}}(\bar{w}_{\text{hold}}) = 0.928$ (oracle): end of descent -0.0066 , -0.0017 (more negative: $\hat{b} > b_{\text{true}}$ over the whole mid range, as $f < 0$ says). $\hat{b}$ estimated from $8/9$ (production): minus locked $8/9$ at the hold start $+0.0014 \pm 0.0009$ (canon), -0.00002 ± 0.00013 (Meng), $\hat{b}$ stays within 0.01 of the seed (0.885 ± 0.017 , 0.896 ± 0.021 at the end), $\sqrt{P_{55}} 0.039 \rightarrow 0.023, 0.027$. Absolute hold level of production with the four blocks: $+0.0012 \pm 0.0016$ (canon), $+0.0004 \pm 0.0023$ (Meng); production as it was (Euler, no mean terms): $+0.019$, $+0.010$. OPEN: a paired hold-slope difference constant- b minus curve of -0.176 ± 0.025 (canon), -0.184 ± 0.015 (Meng) $\times 10^{-6}$ per step; not the b'_{true} cross-moment ($< 0.002 \times 10^{-6}$ on the filter's P , probe_lockb_bprime_line.m); with b locked at the hold value it vanishes on canon (-0.006 ± 0.032) but not on Meng (-0.165 ± 0.014), so it is not simply the slope mismatch at the hold height either; -0.0014 over a 5 s hold, mechanism open)

Jacobians and measurement (the entries that change; code: state writing, $\partial \bar{a}' / \partial \hat{b} = +\bar{a}' / \hat{b}$)

$$\begin{aligned} \frac{\partial \hat{a}_w[k+1]}{\partial \hat{b}[k]} &= \frac{\partial \bar{a}'}{\partial \hat{b}} \Delta \hat{w}[k], \quad \frac{\partial \hat{b}[k+1]}{\partial \hat{b}[k]} = 1 \\ y_2[k] &= \bar{a}_w[k] - \bar{a}' \nabla_d \bar{w}_d[k] + n_a[k], \quad H_{24} = 1 - \frac{\partial \bar{a}'}{\partial \hat{a}_w} \nabla_d \bar{w}_d[k], \quad H_{25} = -\frac{\partial \bar{a}'}{\partial \hat{b}} \nabla_d \bar{w}_d[k] \\ g_n[k] &= (1 - \lambda_c) [0, 0, -1, \bar{a}', 0]^\top, \quad \tilde{Q}_{5j} = \tilde{Q}_{j5} = 0 \end{aligned}$$

(both $\hat{b}$ columns carry a displacement factor and vanish together in a hold, Section 3 of observability-workflow; locked- b arm: the filter's $P_{i5} = 0$, so the added line vanishes and the model-error lines, including the b'_{true} cross-moment, remain)

12 Covariance propagation of the self-referential law: the weight κ

(arms whose slope reads $\hat{a}_w$, Sections 10–11; $A := \partial \bar{a}' / \partial \hat{a}_w = -2\hat{b}(1 - \hat{a}_w)$; $z := e_{\bar{a}_w} + \bar{a}' e_{\delta \bar{w}_3}$ the law-slaved combination of Section 2 of the update-half document; everything else as in Sections 1–11)

Linear error dynamics in the $(e_{\delta \bar{w}_3}, z)$ basis (from Section 10, $e_{\bar{a}_w} = z - \bar{a}' e_{\delta \bar{w}_3}$, $\bar{a}'[k+1] = \bar{a}' + \bar{a}'' \Delta \hat{w}$, $\bar{a}'' = A \bar{a}'$)

$$\begin{aligned} z[k+1] &= (1 + A \Delta \hat{w}) z[k] - A \Delta \hat{w} \bar{a}' e_{\Delta \bar{w}}[k] \\ e_{\delta \bar{w}_3}[k+1] &= (\lambda_c + F_{dw} \bar{a}') e_{\delta \bar{w}_3}[k] + (1 - \lambda_c) e_{\delta \bar{w}_1}[k] - F_{dw} z[k] - \tilde{\varepsilon}_w[k] \\ \ell_{41} + \bar{a}' \ell_{31} &= 0 \iff \text{Cov}(z, e_{\delta \bar{w}_1}) = 0 \end{aligned}$$

(the amplification $1 + A \Delta \hat{w}$ acts on z ; the position noise enters the gain through the law as $\bar{a}' \tilde{\varepsilon}_w$ and is removed by the y_1 update exactly when the second line holds – the $\mathbf{a}'_{\text{true}}$ arm keeps it in every segment, $|\ell_{41} + \bar{a}' \ell_{31}| < 0.002$)

Measured ($\mathbf{b}_{\text{true}}$ arm, four blocks, canon 1.0–1.5 s, 10–30 seeds; Meng 7–10 s in brackets)

$$\begin{aligned} \kappa = 1 : \quad & P_{44} \text{ grows } \times 16 \text{ along the descent} \\ & \ell_{41} = +0.036, \quad -\bar{a}' \ell_{31} = -0.157, \quad \ell_{41} + \bar{a}' \ell_{31} = +0.19 [+0.039] \\ & \sigma(\text{predict}) = 0.025, \quad \sigma(y_1 \text{ leg}) = 0.009, \quad \text{corr} = -0.40 \quad (\mathbf{a}'_{\text{true}} \text{ arm: } 0.018, 0.017, -0.99) \\ & \sigma_{\text{seed}}(\hat{a}_w - \bar{a}_w) = 0.026 [0.032] \text{ at the worst instant,} \quad \sigma_{\text{seed}} / \sqrt{P_{44}} = 0.66 [0.81] \\ \kappa = 0 : \quad & \ell_{41} + \bar{a}' \ell_{31} = 0.002 [0.001], \quad \sigma_{\text{seed}} = 0.0045 [0.0079] \\ & \sigma_{\text{seed}} / \sqrt{P_{44}} = 1.24 [1.60] \text{ (fast),} \quad 1.75 [2.07] \text{ (hold)} \end{aligned}$$

Weighted propagation (covariance only; the state predict, the mean term and H_2 keep A)

$$\tilde{F}_e(4, 4) = 1 + F_{dw} \bar{a}' + \kappa A \Delta \hat{w}, \quad \kappa \in [0, 1]$$

Calibration (seeds 1:10, both trajectories; honesty $= \sigma_{\text{seed}} / \sqrt{P_{44}}$, geometric mean over $\{\text{fast}, \text{hold}\} \times \{\text{canon}, \text{Meng}\}$)

$$\begin{aligned} \kappa &= 0, 0.25, 0.5, 0.75, 1 \\ \text{honesty (geo. mean)} &= 1.64, 1.27, \mathbf{0.96}, 0.81, 0.84 \\ \sigma_{\text{seed}} \text{ worst instant, canon} &= 0.0045, 0.0046, \mathbf{0.0054}, 0.0109, 0.0264 \\ \sigma_{\text{seed}} \text{ worst instant, Meng} &= 0.0079, 0.0101, \mathbf{0.0158}, 0.0249, 0.0323 \\ \text{hold honesty canon / Meng} &= 1.75/2.07, 1.68/1.25, \mathbf{1.19/1.06}, 0.88/1.07, 0.87/1.08 \end{aligned}$$

Validation (seeds 11:20, $\kappa = 0.5$ vs 1)

$$\begin{aligned} \sigma_{\text{seed}} \text{ worst instant} : & 0.0237 \rightarrow 0.0026 \text{ (canon), } 0.0272 \rightarrow 0.0082 \text{ (Meng)} \\ \text{hold } \sigma_{\text{seed}} : & 0.0103 \rightarrow 0.0056, \quad 0.0091 \rightarrow 0.0081 \quad \text{hold honesty : } 1.43 \rightarrow 1.25, \quad 1.36 \rightarrow 1.29 \\ \text{hold level} : & +0.0022 \pm 0.0031 \rightarrow +0.0012 \pm 0.0016, \quad +0.0001 \pm 0.0027 \rightarrow +0.0004 \pm 0.0024 \end{aligned}$$

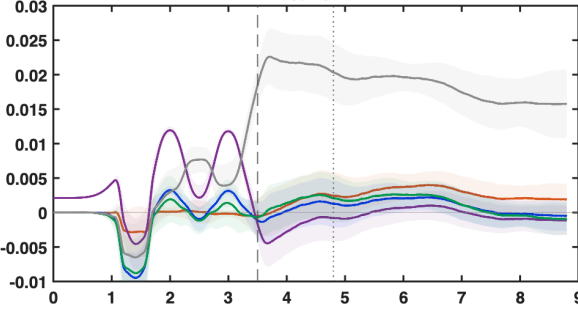
($\kappa = \text{code ctrl_const.fe44_Aa_scale}$. STATUS: $\kappa = 0.5$ is a CALIBRATED value for the $\mathbf{b}_{\text{true}}$ arm only (observable: the honesty ratio; calibration and judgement on disjoint seeds). It was made the production default for a few hours on 2026-09-04 and REVERTED to $\kappa = 1$ the same night: on the

arms with a constant b the smaller P_{44} lets the constant- b model error of Section 11 stand – b locked at $8/9$, canon, 10 seeds: hold -0.0197 ± 0.0007 , every seed at the $\bar{a}$ floor ($+0.0006$ at $\kappa = 1$); $\hat{b}$ estimated (production): hold $+0.0030$ vs $+0.0017$, worst-instant mean -0.031 vs -0.034 (no gain), $\hat{b}$ pulled to 0.852 – 0.858 with $\sqrt{P_{55}}$ collapsing to 0.004 (canon). The calibration does not transfer across arms: the term it weights competes with the y_2 correction of the model error, which the b_{true} arm does not have. What it stands in for: the split of the gain error into a free part, amplified by $1 + A\Delta\hat{w}$, and a law-slaved part that follows $e_{\delta\bar{w}_3}$ and the growth of $\bar{a}'$, which the first-order propagation treats alike; the closed form is open, and a production-safe version must be derived (or calibrated) with the model error present. Not part of this: the y_2 weight – $\text{NIS}_2 = e_{y_2}^2/S_2 = 0.27$ – 0.34 in every segment, both trajectories, both κ (the IF colour design), $\text{NIS}_1 = 1.00$; nothing segment-specific to correct there)

$z - a_z/a_{nom}$: 10-seed mean, 0.5 s window

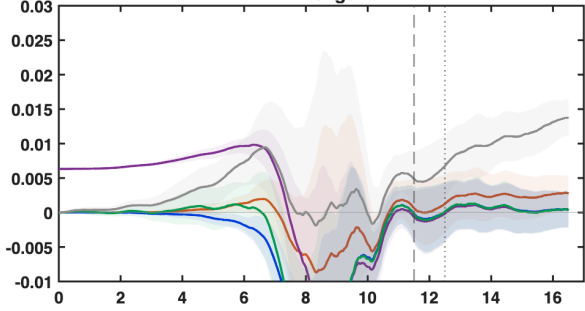
$b_{true}(w)$ curve (A) b locked 8/9 b locked at $b_{true}(w_{hold})$
 b hat estimated (prod) production as it was (Euler)

canon

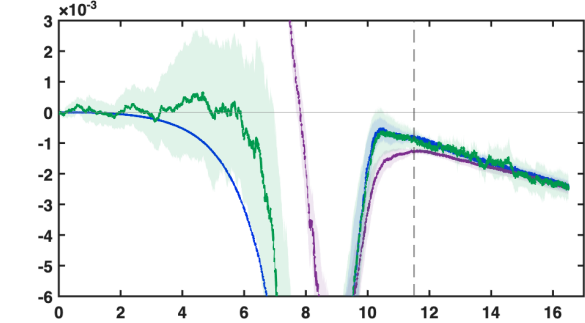
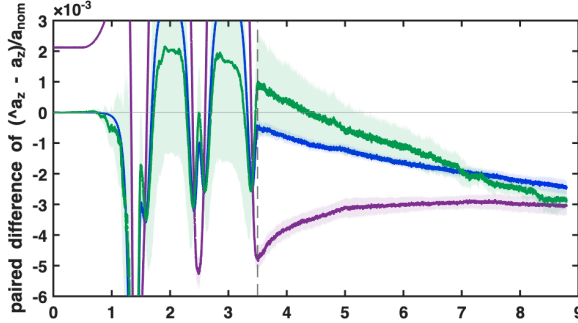


$b_{true}(w)$ curve (A) b locked 8/9 b locked at $b_{true}(w_{hold})$
 b hat estimated (prod) production as it was (Euler)

meng

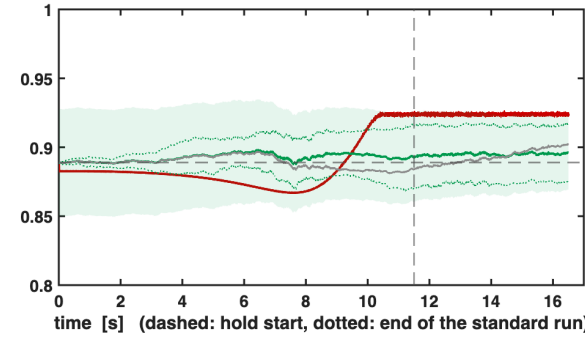
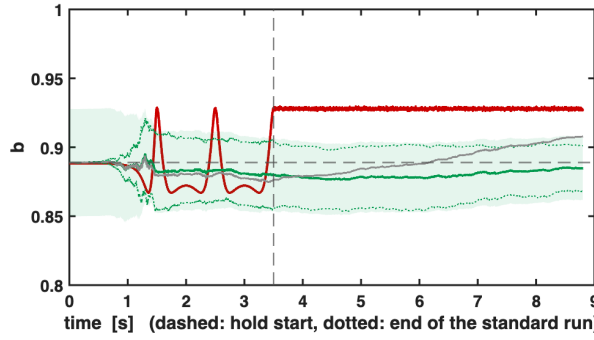


b locked 8/9 minus curve arm b locked at $b_{true}(w_{hold})$ minus curve arm b hat estimated b locked 8/9 minus curve arm b locked at $b_{true}(w_{hold})$ minus curve arm b hat estimated (prod) minus curve arm



$b_{true}(w)$ at the true height $8/9$ (seed, locked value) b prod (10-seed mean; band $\pm \sqrt{P_{ss}}$)
 $b \pm \text{sd over seeds}$ b production as it was

$b_{true}(w)$ at the true height $8/9$ (seed, locked value) b prod (10-seed mean; band $\pm \sqrt{P_{ss}}$)
 $b \pm \text{sd over seeds}$ b production as it was



time [s] (dashed: hold start, dotted: end of the standard run)

time [s] (dashed: hold start, dotted: end of the standard run)

